

# WEBRUM-99: Best Practice Summary

Series: WEBRUM — Web Real User Monitoring | Notebook: 99 | Created: March 2026 | Last Updated: 04/04/2026

## Overview

This notebook consolidates every actionable best practice from the WEBRUM series (notebooks 01-08) into a single definitive reference. Each practice specifies the exact setting, value, or action to take — no hedging, no "it depends." Practices are organized by category and prioritized as Critical, Recommended, or Optional.

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## Prerequisites

| Requirement           | Details                                                            |
|-----------------------|--------------------------------------------------------------------|
| Dynatrace Environment | SaaS with Grail enabled                                            |
| RUM Enabled           | At least one web application with RUM injection active             |
| Permissions           | storage:events:read , storage:metrics:read , storage:entities:read |
| Previous Notebooks    | WEBRUM-01 through WEBRUM-08 (for full context)                     |

## 1. RUM Agent Configuration

| # | Best Practice           | Recommended Setting/Value                                                               | Priority | Source    |
|---|-------------------------|-----------------------------------------------------------------------------------------|----------|-----------|
| 1 | Use automatic injection | Injection method: <b>Automatic</b> (OneAgent modifies HTML responses at the web server) | Critical | WEBRUM-01 |

| # | Best Practice                                   | Recommended Setting/Value                                                                                                         | Priority    | Source               |
|---|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|
| 2 | Filter out bots from all RUM queries            | Always apply <code>filter user.type == "REAL_USER"</code> in every DQL query against <code>user.sessions</code>                   | Critical    | WEBRUM-01            |
| 3 | Use manual injection for CDN/static-hosted apps | Add RUM <code>&lt;script&gt;</code> tag to <code>&lt;head&gt;</code> before all other scripts when no server-side OneAgent exists | Recommended | WEBRUM-02            |
| 4 | Use agentless RUM only as last resort           | Select agentless monitoring only when OneAgent cannot be deployed at all                                                          | Optional    | WEBRUM-02            |
| 5 | Set session timeout to 30 minutes               | Keep the default 30-minute inactivity timeout for session boundaries                                                              | Recommended | WEBRUM-01            |
| 6 | Enable Fetch API monitoring                     | <b>Settings &gt; Web and mobile monitoring &gt; Async web requests and SPAs &gt; Fetch requests: Enabled</b>                      | Critical    | WEBRUM-02            |
| 7 | Deploy RUM + Synthetic together                 | Use synthetic for availability SLAs and baselines; use RUM for real user experience                                               | Critical    | WEBRUM-01, WEBRUM-08 |

## 2. SPA Instrumentation

| #  | Best Practice                                                                              | Recommended Setting/Value                                                                                                                                     | Priority    | Source    |
|----|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 8  | Enable SPA route change capture                                                            | <b>Settings &gt; Web and mobile monitoring &gt; Async web requests and SPAs &gt; SPA route changes: Enabled</b>                                               | Critical    | WEBRUM-02 |
| 9  | Enable Async Web Requests capture                                                          | <b>Settings &gt; Web and mobile monitoring &gt; Async web requests and SPAs &gt; Async web requests: Enabled</b>                                              | Critical    | WEBRUM-02 |
| 10 | Configure user action naming rules for dynamic URLs                                        | Replace dynamic segments: <code>/users/{id}</code> becomes <code>/users/{*}</code> via <b>Settings &gt; Web and mobile monitoring &gt; User action naming</b> | Critical    | WEBRUM-02 |
| 11 | Use <code>data-dtname</code> attribute on interactive elements                             | Add <code>data-dtname="Checkout"</code> to buttons/links for meaningful action names                                                                          | Recommended | WEBRUM-02 |
| 12 | Load RUM agent before the SPA router initializes                                           | Place the RUM <code>&lt;script&gt;</code> tag in <code>&lt;head&gt;</code> before framework bootstrap scripts                                                 | Critical    | WEBRUM-02 |
| 13 | Use <code>dtrum.enterAction()</code> / <code>dtrum.leaveAction()</code> for custom actions | Wrap business-critical interactions in manual RUM actions when auto-detection is insufficient                                                                 | Recommended | WEBRUM-02 |
| 14 | Validate SPA instrumentation after setup                                                   | Query for <code>RouteChange</code> actions — if none appear, SPA monitoring is misconfigured                                                                  | Critical    | WEBRUM-02 |
| 15 | Separate mobile WebView and desktop into distinct RUM apps                                 | Create two Dynatrace applications; use application detection rules based on User-Agent string                                                                 | Recommended | WEBRUM-02 |
| 16 | Add custom User-Agent suffix for WebView detection                                         | Android: <code>setUserAgentString()</code> / iOS: <code>applicationNameForUserAgent</code> with app name + version                                            | Recommended | WEBRUM-02 |
| 17 | Enable hybrid WebView monitoring in mobile SDK                                             | Android: <code>hybridWebView { enabled true; domains "your-domain.com" }</code> / iOS: <code>DTXHybridApplication = true</code>                               | Recommended | WEBRUM-02 |

| #  | Best Practice                         | Recommended Setting/Value                                                                                         | Priority | Source    |
|----|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------|-----------|
| 18 | Never instrument third-party WebViews | Only add your own domains to <code>domains / DTXMonitoredDomains</code> — never OAuth, payment, or external pages | Critical | WEBRUM-02 |

### 3. Core Web Vitals Thresholds

| #  | Best Practice                                                         | Recommended Setting/Value                                                                                   | Priority    | Source    |
|----|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 19 | Target LCP at or below 2.5 seconds                                    | <b>Largest Contentful Paint p75 &lt;= 2500ms.</b> Poor threshold: > 4000ms                                  | Critical    | WEBRUM-03 |
| 20 | Target INP at or below 200 milliseconds                               | <b>Interaction to Next Paint p75 &lt;= 200ms.</b> Poor threshold: > 500ms                                   | Critical    | WEBRUM-03 |
| 21 | Target CLS at or below 0.1                                            | <b>Cumulative Layout Shift p75 &lt;= 0.1.</b> Poor threshold: > 0.25                                        | Critical    | WEBRUM-03 |
| 22 | Use p75 (not average) for CWV evaluation                              | Google uses 75th percentile as the pass/fail threshold for CWV                                              | Critical    | WEBRUM-03 |
| 23 | Track CWV per page, not just per app                                  | Aggregate CWV by <code>action.name</code> to identify which specific pages fail thresholds                  | Recommended | WEBRUM-03 |
| 24 | Set explicit <code>width</code> and <code>height</code> on all images | Prevents CLS from images loading without reserved space                                                     | Critical    | WEBRUM-03 |
| 25 | Use <code>font-display: swap</code> with fallback font metrics        | Prevents CLS from FOUT (Flash of Unstyled Text) during web font loading                                     | Recommended | WEBRUM-03 |
| 26 | Use fixed dimensions on ads, embeds, and iframes                      | Set explicit <code>width / height</code> on all third-party content containers to prevent layout shifts     | Recommended | WEBRUM-03 |
| 27 | Break long JavaScript tasks to improve INP                            | Split tasks > 50ms using <code>requestIdleCallback</code> , <code>setTimeout(0)</code> , or web workers     | Critical    | WEBRUM-03 |
| 28 | Batch DOM reads and writes to reduce INP                              | Avoid forced layout/reflow by separating DOM read and write operations                                      | Recommended | WEBRUM-03 |
| 29 | Build a CWV scorecard query                                           | Use the single-query pattern from WEBRUM-03 to show % Good/NI/Poor for all three metrics                    | Recommended | WEBRUM-03 |
| 30 | Track CWV trends at hourly (LCP/INP) and daily (CLS) granularity      | Use <code>makeTimeseries</code> with <code>interval:1h</code> for LCP/INP, <code>interval:1d</code> for CLS | Recommended | WEBRUM-03 |

### 4. Performance Optimization

| #  | Best Practice                               | Recommended Setting/Value                                                                        | Priority    | Source    |
|----|---------------------------------------------|--------------------------------------------------------------------------------------------------|-------------|-----------|
| 31 | Target TTFB at or below 800 milliseconds    | <b>Time to First Byte p75 &lt;= 800ms</b> (Google recommendation)                                | Critical    | WEBRUM-06 |
| 32 | Use DNS prefetching for third-party domains | Add <code>&lt;link rel="dns-prefetch" href="//cdn.example.com"&gt;</code> for external resources | Recommended | WEBRUM-06 |
| 33 | Enable HTTP/2 on all web servers            | Reduces TCP connection overhead; multiplexes requests on a single connection                     | Critical    | WEBRUM-06 |

| #  | Best Practice                                 | Recommended Setting/Value                                                                                                          | Priority    | Source    |
|----|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 34 | Enable TLS 1.3 with OCSP stapling             | Reduces SSL handshake time by one round trip                                                                                       | Recommended | WEBRUM-06 |
| 35 | Defer non-critical JavaScript                 | Use <code>defer</code> or <code>async</code> attributes on <code>&lt;script&gt;</code> tags that are not needed for initial render | Critical    | WEBRUM-06 |
| 36 | Lazy-load below-the-fold images               | Use <code>loading="lazy"</code> on <code>&lt;img&gt;</code> tags not visible in the initial viewport                               | Recommended | WEBRUM-06 |
| 37 | Monitor the DOM Interactive to Load Event gap | A large gap indicates excessive sub-resource loading; target < 2 seconds                                                           | Recommended | WEBRUM-06 |
| 38 | Deploy CDN edge servers for high-TTFB regions | If TTFB is high for specific geographies, add CDN nodes closer to those users                                                      | Critical    | WEBRUM-06 |
| 39 | Prioritize slow pages by impact score         | Calculate <code>impact_score = page_views * avg_duration_ms</code> and fix highest-impact pages first                              | Recommended | WEBRUM-06 |
| 40 | Set page load performance SLA at 3 seconds    | Track % of page loads with duration <= 3s as primary performance SLA                                                               | Critical    | WEBRUM-08 |

## 5. Error Monitoring

| #  | Best Practice                                                   | Recommended Setting/Value                                                                                                 | Priority    | Source               |
|----|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------|----------------------|
| 41 | Target error session rate below 2%                              | <b>% of sessions with <code>error.count &gt; 0</code> &lt; 2%</b>                                                         | Critical    | WEBRUM-05, WEBRUM-08 |
| 42 | Rank errors by affected session count, not occurrence count     | A retry loop in one session inflates occurrence count; <code>countDistinct(session.id)</code> is the true impact metric   | Critical    | WEBRUM-05            |
| 43 | Monitor XHR/fetch errors separately from JS errors              | Filter <code>error.type == "XHR_ERROR"</code> or <code>error.type == "FETCH_ERROR"</code> to isolate backend API failures | Recommended | WEBRUM-05            |
| 44 | Enable rage click detection                                     | Dynatrace captures rage clicks automatically; query <code>rage.click == true</code> to find frustrated users              | Critical    | WEBRUM-05            |
| 45 | Correlate errors with session outcomes                          | Compare bounce rate and avg actions for "With Errors" vs "No Errors" sessions to quantify error impact                    | Recommended | WEBRUM-05            |
| 46 | Segment error rates by browser family                           | Query <code>error.count &gt; 0</code> grouped by <code>browser.family</code> to identify browser-specific issues          | Recommended | WEBRUM-05            |
| 47 | Alert on new error messages                                     | Query errors with <code>min(timestamp)</code> in the last hour to detect errors introduced by recent deployments          | Critical    | WEBRUM-08            |
| 48 | Use <code>dtrum.reportError()</code> for custom business errors | Report application-specific errors (e.g., payment validation failures) via the RUM API                                    | Recommended | WEBRUM-05            |

## 6. Session Replay and Privacy

| #  | Best Practice                                                         | Recommended Setting/Value                                                                                                                      | Priority    | Source    |
|----|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 49 | Set Session Replay default masking to "Mask user input"               | <b>Settings &gt; Session Replay &gt; Data privacy &gt; Default masking level: Mask user input</b>                                              | Critical    | WEBRUM-07 |
| 50 | Block entire PII containers with <code>data-dtrum-block="true"</code> | Add <code>data-dtrum-block="true"</code> to any <code>&lt;div&gt;</code> containing sensitive data (SSN, CC, health info)                      | Critical    | WEBRUM-07 |
| 51 | Add CSS masking rules for all form inputs with sensitive data         | <b>Settings &gt; Session Replay &gt; Data privacy:</b> add selectors like <code>.credit-card-form input</code> , <code>[data-sensitive]</code> | Critical    | WEBRUM-07 |
| 52 | Set replay sampling rate based on traffic volume                      | High-traffic production: <b>1-5%</b> ; internal apps: <b>10-25%</b> ; pre-prod: <b>50-100%</b>                                                 | Recommended | WEBRUM-07 |
| 53 | Enable error-triggered replay capture                                 | Configure conditional capture: automatically record sessions when JavaScript errors occur                                                      | Recommended | WEBRUM-07 |
| 54 | Enable frustration-triggered replay capture                           | Configure conditional capture for sessions with rage clicks or exit intent                                                                     | Recommended | WEBRUM-07 |
| 55 | Err on over-masking, not under-masking                                | It is better to mask non-sensitive content than to accidentally capture PII                                                                    | Critical    | WEBRUM-07 |
| 56 | Conduct a privacy review before enabling Session Replay in production | Review all captured content against GDPR, CCPA, and HIPAA requirements before go-live                                                          | Critical    | WEBRUM-07 |
| 57 | Use Dynatrace native replay over third-party tools                    | Native replay integrates with performance waterfall, backend traces, and Dynatrace Intelligence; third-party tools lack this correlation       | Recommended | WEBRUM-07 |
| 58 | Follow the replay investigation workflow                              | 1) Identify via DQL, 2) Watch replay, 3) Correlate with waterfall, 4) Note pattern, 5) Quantify with DQL                                       | Recommended | WEBRUM-07 |

## 7. Session Analysis

| #  | Best Practice                                         | Recommended Setting/Value                                                                                                                                                                                                                      | Priority    | Source                  |
|----|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------|
| 59 | Define custom session properties for business context | Configure <b>Settings &gt; Web and mobile monitoring &gt; Session and user action properties</b> with:<br><code>session.user_id</code> , <code>session.plan_type</code> ,<br><code>session.cart_value</code> , <code>session.ab_variant</code> | Critical    | WEBRUM-04               |
| 60 | Target bounce rate below 40%                          | <b>% of sessions with action.count == 1 &lt; 40%</b>                                                                                                                                                                                           | Recommended | WEBRUM-04,<br>WEBRUM-08 |
| 61 | Segment sessions by engagement level                  | Bucket sessions: Bounce (1 action), Low (2-3), Medium (4-10), High (10+)                                                                                                                                                                       | Recommended | WEBRUM-04               |
| 62 | Track entry and exit pages                            | Use <code>takeFirst(action.name)</code> and <code>takeLast(action.name)</code> grouped by <code>session.id</code> to map user journeys                                                                                                         | Recommended | WEBRUM-04               |

| #  | Best Practice                                                            | Recommended Setting/Value                                                                                                                                            | Priority    | Source                  |
|----|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------|
| 63 | Use session properties for conversion tracking, not URL pattern matching | Mark conversions via session properties instead of fragile <code>action.name ~ "*confirmation*"</code> patterns                                                      | Critical    | WEBRUM-04               |
| 64 | Analyze sessions by hour of day for capacity planning                    | Group sessions by <code>getHour(timestamp)</code> over 7 days to identify peak traffic windows                                                                       | Optional    | WEBRUM-04               |
| 65 | Segment by geography and device                                          | Break down sessions by <code>country</code> , <code>browser.family</code> , <code>os.family</code> , and <code>connection.type</code> to find regional/device issues | Recommended | WEBRUM-04,<br>WEBRUM-06 |

## 8. Dashboards and Alerting

| #  | Best Practice                                                                        | Recommended Setting/Value                                                                                          | Priority    | Source    |
|----|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 66 | Set Apdex threshold T = 3 seconds for page loads                                     | Satisfied: $\leq 3s$ , Tolerating: 3-12s, Frustrated: $> 12s$                                                      | Critical    | WEBRUM-08 |
| 67 | Set Apdex threshold T = 2.5 seconds for XHR and route changes                        | Satisfied: $\leq 2.5s$ , Tolerating: 2.5-10s, Frustrated: $> 10s$                                                  | Critical    | WEBRUM-08 |
| 68 | Target Apdex score above 0.85                                                        | Apdex 0.85-0.93 = Good; 0.94-1.00 = Excellent; below 0.70 = investigate immediately                                | Critical    | WEBRUM-08 |
| 69 | Build an executive dashboard with 6 KPIs                                             | Tiles: Apdex, session count, error rate %, bounce rate %, CWV pass rate %, avg session duration                    | Recommended | WEBRUM-08 |
| 70 | Build an operational dashboard with real-time error view                             | Tiles: error count per 15m by type, active error table (last 1h), performance SLA % under 3s                       | Recommended | WEBRUM-08 |
| 71 | Alert on error rate spike                                                            | Threshold: <b><math>&gt; 5\%</math> of sessions with errors sustained for 15 minutes</b>                           | Critical    | WEBRUM-08 |
| 72 | Alert on Apdex drop                                                                  | Threshold: <b>Apdex <math>&lt; 0.7</math> sustained for 15 minutes</b>                                             | Critical    | WEBRUM-08 |
| 73 | Alert on performance degradation                                                     | Threshold: <b>p75 page load <math>&gt; 5</math> seconds sustained for 15 minutes</b>                               | Critical    | WEBRUM-08 |
| 74 | Alert on traffic anomaly                                                             | Threshold: <b>session count <math>&lt; 50\%</math> of previous day's hourly average</b>                            | Recommended | WEBRUM-08 |
| 75 | Alert on CWV regression                                                              | Threshold: <b>LCP p75 <math>&gt; 4</math> seconds sustained for 30 minutes</b>                                     | Recommended | WEBRUM-08 |
| 76 | Use metric event evaluation window of 15 minutes                                     | <b>Settings &gt; Anomaly detection &gt; Metric events:</b> evaluation window <b>15m</b> , sliding window <b>5m</b> | Recommended | WEBRUM-08 |
| 77 | Prefer Dynatrace Intelligence anomaly detection over static thresholds in production | Dynatrace Intelligence automatically baselines normal behavior, reducing false positives from seasonal patterns    | Recommended | WEBRUM-08 |

| #  | Best Practice                                   | Recommended Setting/Value                                                                                      | Priority    | Source    |
|----|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 78 | Combine RUM and synthetic in a single dashboard | Use <code>append</code> to show side-by-side comparison: RUM real-user metrics vs synthetic clean-room metrics | Recommended | WEBRUM-08 |
| 79 | Track CWV pass rate as executive KPI            | Target: <b>&gt; 75% of page loads meeting all three CWV thresholds</b>                                         | Recommended | WEBRUM-08 |

## 9. DQL Query Patterns

| #  | Best Practice                                                           | Recommended Setting/Value                                                                                            | Priority    | Source               |
|----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------|----------------------|
| 80 | Always include <code>from: time</code> range on RUM queries             | Use <code>from:-1h</code> for real-time, <code>from:-24h</code> for daily, <code>from:-7d</code> for trends          | Critical    | WEBRUM-01            |
| 81 | Always filter <code>user.type == "REAL_USER"</code> on session queries  | Excludes bots ( <code>ROBOT</code> ) and synthetic monitors ( <code>SYNTHETIC</code> ) from real-user analysis       | Critical    | WEBRUM-01            |
| 82 | Use <code>isNotNull()</code> before aggregating optional fields         | Apply filter <code>isNotNull(web_vitals.largest_contentful_paint)</code> before CWV aggregations                     | Critical    | WEBRUM-03            |
| 83 | Use <code>countDistinct(session.id)</code> for error impact measurement | Never use <code>count()</code> alone — it inflates impact from retry loops in single sessions                        | Critical    | WEBRUM-05            |
| 84 | Require minimum sample size before drawing conclusions                  | Use filter <code>page_views &gt; 20</code> (or similar) to avoid misleading statistics from low-volume pages         | Recommended | WEBRUM-03, WEBRUM-06 |
| 85 | Use nanosecond-to-millisecond conversion: <code>/ 1000000.0</code>      | Dynatrace duration fields are in nanoseconds; divide by 1,000,000 for milliseconds, 1,000,000,000 for seconds        | Critical    | WEBRUM-03            |
| 86 | Alias all aggregations before using in <code>sort</code>                | Always write <code>summarize c = count()</code> then <code>sort c desc</code> , never <code>sort count() desc</code> | Critical    | All                  |

*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*